RACE
Regular Analysis Through Continuous Exercise

Chemical Kinetics : RACE-01

Rate of reaction

1. In the formation of sulphur trioxide by the contact process $2SO_2(g) + O_2(g) \longrightarrow 2SO_3(g)$; the rate of reaction is expressed as $-\frac{d[O_2]}{dt} = 2.5 \times 10^{-4}$ mol L^{-1} s^{-1}

The rate of disappearance of (SO_2) will be

(A) 5×10^{-4} mol L^{-1} s^{-1} (B) -2.25×10^{-4} mol L^{-1} s^{-1}

(C) 3.75×10^{-4} mol L^{-1} s^{-1} (D) 50.0×10^{-4} mol L^{-1} s^{-1}

CCK085

2. If the concentration of the reactants in the reaction $2A + B \rightarrow C + D$ is increased by three folds, the rate of the reaction will be increased by-

(A) 27 times (B) 9 times

(C) 64 times (D) 01 times

CCK185

First order reactions

3. If decomposition of H_2O_2 in aqueous medium follows first order kinetics then which of the following are correct

(A) It is an example of single step reaction (B) It is an example of complex reaction

(C) Half life of reaction varies with time (D) Half life of reaction varies with temperature

CCK186

4. Substance A reacts according to a first order rate law with $k = 5.0 \times 10^{-5}$ s^{-1}. If the initial conc. of A is 1.0 M, the initial rate is

(A) 1×10^{-5} Ms^{-1} (B) 5.0×10^{-5} Ms^{-1}

(C) 1×10^{-4} Ms^{-1} (D) 5.0×10^{-4} Ms^{-1}

CCK187

5. The half-life period of a first order reaction is 100 seconds. Its rate constant is $x \times 10^{-3}$. Find the value of x:

CCK188

6. In the equation of a first order reaction $C_t = C_0e^{-kt}$ if the initial concentration C_0 is increased m times then

(A) The value of k will increase m times (B) The value of k will decrease m times

(C) The value of k will remains unchanged (D) None of these

CCK189

7. 75% of a first order reaction was completed in 32 minutes. When was 50% of the reaction completed.

(A) 24 minutes (B) 8 minutes

(C) 16 minutes (D) 4 minutes

CCK190

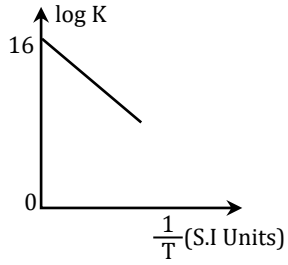
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8. Half lives for a zero order (A → B) & 1st order reaction (A → P) are same. Then starting with same initial concentration of 'A' the ratio of initial rate of first order & zero order will be -

(A) 0.693 (B) $\frac{0.693}{2}$ (C) 0.693 × 2 (D) $\frac{1}{0.693}$

CCK191

9. For any fist order reaction following observation is made



If at temperature (T) half life of the reaction is 6930 sec. & at temperature (T') half life of the reaction is 0.693 msec. Then calculate $\frac{T'}{T}$.

(A) 2 (B) 4 (C) 6 (D) 8

CCK192

Pseudo first order reaction

10. The rate law for the reaction : $2\,C + D \rightarrow A + E$ is $-\frac{d[D]}{dt} = k[C]^2[D]$

If C is present in large excess, the order of the reaction will be -

(A) Zero (B) First (C) Second (D) Third

CCK193

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	A	A	BD	B	6.93	C	C	C	A	B

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Chemical Kinetics : RACE-02

Second order kinetics; n^{th} order kinetics

1. The half life period of a chemical reaction is given by $t_{\frac{1}{2}} : \propto \frac{1}{a^{n-1}}$

The order of the reaction is :

(A) n (B) n – 1

(C) n + 1 (D) None of these

CCK194

2. In a second order reaction 20% of a substance is dissociated in 40 minutes. The time taken by the 80% of its dissociation is

(A) 160 minutes (B) 640 minutes

(C) 200 minute (D) 320 minute

CCK195

Calculation of First order rate constant using Pressure; Volume; Optical Rotation

3. For any reaction A(g) → P(s) following observation is made :

Time (min)	**Partial pressure of A (in bar)**
0	20
20	10
60	5

Order of the reaction may be -

(A) –1 (B) 0 (C) 1 (D) 2

CCK196

Experimental determination of order of reaction

4. Following data are obtained for the reaction A + B → product

[A] **(mole L^{-1})**	**[B]** **(mole L^{-1})**	**Rate** **(mole L^{-1} s^{-1})**
4×10^{-2}	4×10^{-2}	2×10^{-2}
4×10^{-2}	8×10^{-2}	4×10^{-2}
2×10^{-2}	8×10^{-2}	2×10^{-2}

The reaction confirms the rate law

(A) $-\frac{d[B]}{dt} = k[A]$ (B) $-\frac{d[B]}{dt} = k[A][B]^2$

(C) $-\frac{d[B]}{dt} = k[A][B]$ (D) $-\frac{d[A]}{dt} = k[B]$

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5. The following kinetic data are provided for a reaction between A and B :

Concentration of A(M)	**Concentration of B(M)**	**Rate of reaction ($M\ min^{-1}$)**
0.50	0.02	1.15×10^{-4}
0.50	0.04	2.30×10^{-4}
0.01	1.00	2.30×10^{-6}
0.02	1.00	0.92×10^{-5}

Then value of the rate constant for the above reaction is equal to :

(A) 1.15×10^{-4} $dm^3/mol\ min$ (B) 2.30×10^{-4} $dm^6/mol^2\ min$

(C) 2.30×10^{-2} $dm^6/mol^2\ min$ (D) 1.15×10^{-2} $dm^3/mol\ min$

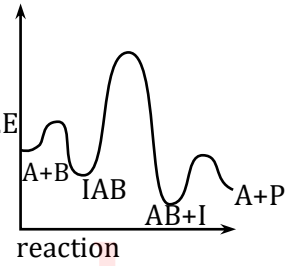
CCK198

Mechanism of Reaction

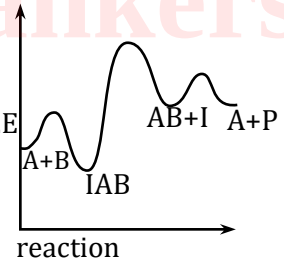
6. The following mechanism has been proposed for the exothermic catalyzed complex reaction.

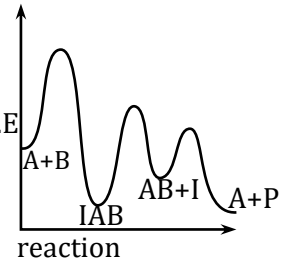
$$A + B \underset{}{\overset{\text{fast}}{\rightleftharpoons}} IAB \xrightarrow{k_1} AB + I \xrightarrow{k_2} P + A$$

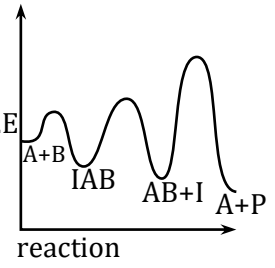
If k_1 is much smaller than k_2. The most suitable qualitative plot of potential energy (P.E.) versus reaction coordinate for the above reaction.

(A)


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(B)


(C)


(D)


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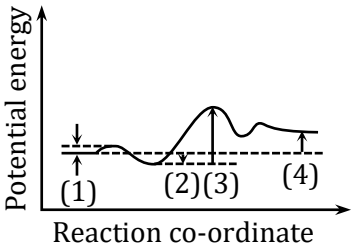
7. Choose the correct set of identifications for the reaction.

Substrate(S) $\xrightarrow{\text{Enzyme(E)}}$ Product (P)

whose mechanism is $E + S \rightleftharpoons ES$

$ES \rightleftharpoons EP$

$EP \rightleftharpoons E + P$



(A) (1) ΔE for $E + S \rightarrow ES$, (2) E_a for $ES \rightarrow EP$, (3) $\Delta E_{overall}$ for $S \rightarrow P$, (4) E_a for $EP \rightarrow E + P$

(B) (1) E_a for $E + S \rightarrow ES$, (2) ΔE for $E + S \rightarrow ES$, (3) E_a for $ES \rightarrow EP$, (4) $\Delta E_{overall}$ for $S \rightarrow P$

(C) (1) E_a for $ES \rightarrow EP$, (2) E_a for $EP \rightarrow E + P$, (3) $\Delta E_{overall}$ for $S \rightarrow P$, (4) ΔE for $EP \rightarrow E + P$

(D) (1) E_a for $E + S \rightarrow ES$, (2) E_a for $ES \rightarrow EP$, (3) E_a for $EP \rightarrow E + P$, (4) $\Delta E_{overall}$ for $S \rightarrow P$

CCK119

8. For an elementary chemical reaction,

$A_2 \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} 2A$, the expression for $\frac{d[A]}{dt}$ is :

(A) $2k_1[A_2] - k_{-1}[A]^2$ (B) $k_1[A_2] - k_{-1}[A]^2$

(C) $2k_1[A_2] - 2k_{-1}[A]^2$ (D) $k_1[A_2] + k_{-1}[A]^2$

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CCK126

ANSWER KEY

Question	1	2	3	4	5	6	7	8
Answer	A	B	D	C	C	A	B	C

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RACE Regular Analysis Through Continuous Exercise

Chemical Kinetics : RACE-03

Arrhenius equation

1. The activation energy of the reaction, A + B ⟶ C + D + 38 kcal is 20 kcal. What would be the activation energy of the following reaction.

C + D ⟶ A + B

(A) 20 kcal (B) –20 kcal

(C) 18 kcal (D) 58 kcal

CCK128

2. A reaction rate constant is given by :

$k = 1.2 \times 10^{14}\, e^{-25000/RT}\ sec^{-1}$.

It means :-

(A) log k versus log T will give a straight line with a slope as 25000

(B) log k versus log T will give a straight line with a slope as –2500

(C) log k versus T will give a straight line with a slope as –25000

(D) log k versus 1/T will give a straight line

CCK135

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3. For a reaction $E_a = 0$ and $k = 3.2 \times 10^8 s^{-1}$ at 300K. The value of frequency factor at 400 K would be :

(A) $6.4 \times 10^4 s^{-1}$ (B) $3.2 \times 10^4 s^{-1}$

(C) $3.2 \times 10^8 s^{-1}$ (D) $3.2 \times 10^5 s^{-1}$

CCK137

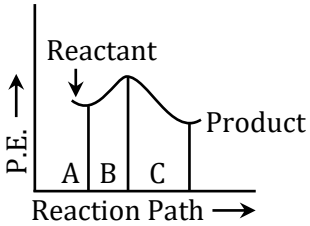
4. In a reaction carried out at 400 K, 0.0001% of the total number of molecules are in activated state. The energy of activation of reaction is -

(A) 0 (B) 7.37 Kcal/mol

(C) 9.21 Kcal/mol (D) 11.05 Kcal/mol

CCK139

5. With respect to the figure given below which of the following statement is correct ?



(A) E_a for the forward reaction is C – B

(B) E_a for the forward reaction is B – A

(C) $E_{a(forward)} > E_{a(backward)}$

(D) E_a(for reverse reaction) = C – A

CCK141

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6. **Statement-1** : In a reversible endothermic reaction, (E_{act}) of forward reaction is higher than that of backward reaction

Statement-2 : The threshold energy of forward reaction is more than that of backward reaction

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

CCK143

7. The energy of activation for a reaction at 27°C is 10 kJ/mol. The presence of catalyst lowers the energy of activation by 75%. Find the factor by which rate of reaction increases at 27°C due to catalyst. **(Take R = 25/3 J/mol-k) [e = 2.7]**

CCK145

8. The slope of the line for the graph of log k versus $\frac{1}{T}$ for the reaction, $N_2O_5 \rightarrow 2NO_2 + \frac{1}{2} O_2$ is – 5000. Calculate the energy of activation of the reaction - (kJ K^{-1} mol^{-1})

(A) 95.7 (B) 9.57 (C) 957 (D) None

CCK199

9. For a reaction, the rate constant is expressed as, $k = A.e^{-40000/T}$. The energy of the activation is :

(A) 40000 cal (B) 88000 cal (C) 80000 cal (D) 8000 cal

CCK200

10. A reaction takes place in three steps. The rate constants are k_1, k_2 and k_3. The overall rate constant $k = \frac{k_1 k_3}{k_2}$. If (energy of activation) E_1, E_2 and E_3 are 60, 30 and 10 kJ. The overall energy of activation is (in kJ).

CCK201

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	D	D	C	D	B	C	19.68	A	C	40

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